

# SkillClaw: Let Skills Evolve Collectively with Agentic Evolver

Collective Skill Evolution for Multi-User LLM Agent Ecosystems

From static skills to a closed-loop, evidence-driven evolution

Evidence

Attribution

Evolution

Distribution

Source: arXiv:2604.08377 (<https://arxiv.org/abs/2604.08377>)

# Agent Skills Remain Static After Deployment — Users Rediscover Solution

## Static paradigm today

- Skills manually installed from a centralized hub
- No runtime adaptation after deployment
- Failures and fixes repeat across users
- Knowledge stays session-local, not ecosystem-level

## Why this matters

- Recurring tasks trigger the same failure/recovery patterns
- Each user independently rediscovers solutions
- Memory-only methods remain instance-bound
- Need a collective, continuously improving skill loop

# Closed-Loop Pipeline: From Multi-User Interaction to Shared Skill Evolution

Multi-user interaction trajectories are collected, analyzed, and used to evolve shared skills that are synchronized back to all users.



Evidence: mine successes/failures across sessions

Attribution: diagnose root causes and boundary conditions

Evolution: generate candidate skill mutations

Distribution: validator-approved skills sync to shared pool

# Agentic Evolver: Evidence → Attribution → Evolution in Three Stages

## 1) Evidence Analysis

Aggregate trajectory fragments to identify where a skill succeeds, fails, or degrades under context shifts.

## 2) Attribution

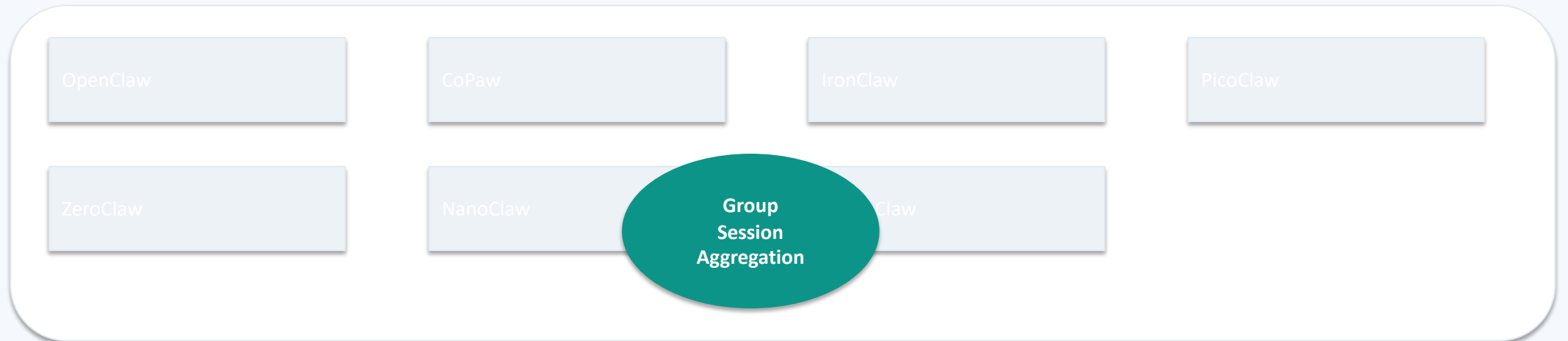
Perform causal diagnosis on failures to separate root causes from incidental artifacts.

## 3) Evolution

Propose skill mutations via open-ended reasoning rather than predefined update rules.

# Cross-User Trajectory Aggregation Exposes Skill Behavioral Boundaries

Compatible systems contribute traces: OpenClaw, CoPaw, IronClaw, PicoClaw, ZeroClaw, NanoClaw, NemoClaw.  
Shared-skill usage across heterogeneous contexts yields complementary success/failure evidence.  
Single-user evidence is often insufficient to distinguish generalizable updates from idiosyncratic fixes.



# Case Study: Multimodal Creative Pipeline Skill — 6 Days of Evolution Atte

Conservative verification outcome: 1/6 candidate evolutions accepted; 5 rejected to prevent regressions.

| Date  | Outcome  | Brief rationale                                          |
|-------|----------|----------------------------------------------------------|
| Day 1 | Rejected | Insufficient evidence for stable gain across contexts.   |
| Day 2 | Rejected | Candidate overfit to a narrow prompt style.              |
| Day 3 | Accepted | Improved robustness without harming prior success cases. |
| Day 4 | Rejected | Introduced regressions in downstream multimodal steps.   |
| Day 5 | Rejected | No consistent improvement over current skill version.    |
| Day 6 | Rejected | Validator detected instability under diverse inputs.     |

# Case Study: Git Push Auth-Fallback Skill — Iterative Refinement Over 6 Days

Iterative refinement outcome: 3/6 accepted updates before performance plateaued.

| Date  | Outcome  | Brief rationale                                                |
|-------|----------|----------------------------------------------------------------|
| Day 1 | Accepted | Added reliable auth fallback path for common push failures.    |
| Day 2 | Rejected | Change duplicated existing logic without measurable gain.      |
| Day 3 | Accepted | Improved error discrimination between auth and network faults. |
| Day 4 | Rejected | Edge-case handling increased complexity with no net benefit.   |
| Day 5 | Accepted | Refined retry sequencing improved successful push completion.  |
| Day 6 | Rejected | No further improvement; signs of convergence/plateau.          |

# Key Takeaways: Three Properties That Distinguish SkillClaw from Existing S

## Collective evolution

Individual interactions contribute to a shared skill ecosystem that improves over time.

## Fully automatic

Runtime interactions drive skill updates without manual curation or explicit user intervention.

## Agentic evolution paradigm

Open-ended reasoning over evidence enables context-aware skill mutation beyond fixed rules.

**Implication: continuously improving agent systems should treat skill evolution as a first-class, closed-loop systems capability.**